

RESEARCH INTERESTS

Programming Languages, Compiler, Formal Verification, Parallelism, and WebAssembly

EDUCATION

Carnegie Mellon University

Pittsburgh, USA

Ph.D. in Software Engineering, advised by Limin Jia

Aug 2023 – Present

Pohang University of Science and Technology (POSTECH)

Pohang, Korea

Bachelor of Computer Science and Engineering & Mathematics (double major)

Feb 2018 – Feb 2023

- GPA : 4.26/4.3 (ranked 1st in class)
- Entered with presidential scholarship offered to one outstanding high school student in the year, and graduated with the highest GPA
- Early graduation by one-year (leave of absence for military duty during Jan 2020 – Dec 2021)

Korea Science Academy of KAIST (*High School for Science-Gifted Students*)

Busan, Korea

Graduated

Feb 2015 – Feb 2018

- GPA : 4.23/4.3 (ranked 1st in class)

ONGOING RESEARCH PROJECTS

MLIR Dialect for WebAssembly

May 2024 – Present

Compilation pipeline for deploying ML applications on various platforms

- Developing MLIR dialect for WebAssembly and dialect conversion passes with formal correctness proof

Hybrid CPU-GPU Task Parallelism

Dec 2023 – Present

Language support and task scheduling techniques for hybrid CPU-GPU task parallelism in MaPLe, SML-like functional language for multicore parallelism

- Developing multi-GPU support and task scheduling techniques to minimize overhead and maximize throughput

PUBLICATIONS

GraFeyn: Efficient Parallel Sparse Simulation of Quantum Circuits, Sam Westrick, Pengyu Liu, [Byeongjee Kang](#), Colin McDonald, Mike Rainey, Mingkuan Xu, Jatin Arora, Yongshan Ding and Umut Acar, IEEE International Conference on Quantum Computing and Engineering (QCE) 2024 (**Best Paper**) ([pdf](#))

Narrowing and Heuristic Search for Symbolic Reachability Analysis of Concurrent Object-Oriented Systems, [Byeongjee Kang](#) and Kyungmin Bae, Science of Computer Programming 2024 (doi.org/10.1016/j.scico.2024.103097)

Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search, [Byeongjee Kang](#) and Kyungmin Bae, International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS) 2022 (doi.acm.org?doi=3563822.3568017)

PUBLICATIONS UNDER REVIEW & PREPRINTS

Fray: An Efficient General-Purpose Concurrency Testing Platform for the JVM, Ao Li, Byeongjee Kang, Vasudev Vikram, Isabella Laybourn, Samvid Dharanikota, Shrey Tiwari, and Rohan Padhye, International Conference on Software Engineering (ICSE) 2025

Mr3: An Execution Framework for Hive and Spark Sungwoo Park, Seunggon Namgung, Byeongjee Kang, Journal of Distributed and Parallel Computing (dx.doi.org/10.2139/ssrn.4587845)

WORK EXPERIENCES

Kodebox Seoul, Korea
Software engineer as military service alternative *Jan 2020 – Dec 2021*

- **CodeChain Foundry**: Open-source blockchain engine based on composable module system
 - Worked as core developer of consensus engine
 - Implemented BLS signature aggregation scheme to optimize block seal size
 - Designed interface for modules to communicate with consensus engine
 - Tech Stacks: Rust, Docker, Kubernetes, Google Cloud Platform
- **Zuzu**: Web-based captable management software
 - Worked as full-stack software engineer
 - Tech Stacks: Django (Python), React (TypeScript), GraphQL, Kubernetes, Google Cloud Platform

HONORS AND SCHOLARSHIPS

ILJU Fellowship (ILJU Academy and Culture Foundation) *Aug 2023 – July 2028*

- Received total of USD 12,000 over eight semesters during graduate studies

Program for Highly Dedicated Students (POSTECH) *Feb 2018 – Feb 2023*

- Presidential scholarship and supporting program in POSTECH offered to **one outstanding freshman** each year

Presidential Science Scholarship (Ministry of Education of Korea) *Feb 2018 – Feb 2023*

- National science scholarship offered to top freshmen students in the field of science and technology

Global Leadership Program (CSE Department, POSTECH) *Sep 2019 – Present*

- Scholarship and research support program for selected students in computer science department

Domun Scholarship (POSTECH) *Feb 2019*

- Scholarship for freshmen with highest GPA

TEACHING EXPERIENCES

Student Mentoring Program (CSE Department, POSTECH) *Feb 2022 – June 2022*

- Served as tutor for Algorithm course

Intensive Research Program (POSTECH) *Jan 2019*

- Served as mentor in research experience program for science high school students

LANGUAGES

Korean (native), English (fluent)